

KEMI AJOS, Finland

WMO: 028620

Lat: 65.673N

Lon: 24.515E

Elev: 3

StdP: 101.29

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.0	-22.4	-27.9	0.3	-24.6	-24.9	0.4	-22.3	16.7	-0.2	15.4	-0.5	4.0	120	0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	23.7	18.2	21.8	17.6	20.2	16.6	19.6	22.7	18.2	20.9	17.1	19.6	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.2	21.4	17.0	12.1	20.0	15.9	11.3	18.5	55.9	22.9	51.6	20.9	48.0	19.8	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.2	13.4	11.9	-27.1	26.1	3.5	2.5	-29.6	28.0	-31.7	29.5	-33.6	30.9	-36.1	32.7		
			-27.3	20.2	3.4	1.7	-29.8	21.4	-31.8	22.5	-33.7	23.4	-36.2	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-8.9	-9.4	-6.1	-0.2	6.7	12.6	16.6	15.2	10.6	4.1	-0.9	-4.3	(6)
(7)		DBStd	9.99	6.54	7.10	5.19	3.13	4.22	3.09	2.96	2.81	2.95	3.68	4.59	5.64	(7)
(8)		HDD10.0	3044	586	543	498	304	121	9	0	1	28	184	328	442	(8)
(9)		HDD18.3	5591	844	777	756	554	361	173	71	104	231	442	578	700	(9)
(10)		CDD10.0	520	0	0	0	0	19	88	205	161	47	1	0	0	(10)
(11)		CDD18.3	26	0	0	0	0	1	2	17	6	0	0	0	0	(11)
(12)		CDH23.3	71	0	0	0	0	3	7	53	8	0	0	0	0	(12)
(13)		CDH26.7	7	0	0	0	0	0	1	7	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	5.9	5.9	5.9	5.3	5.3	5.4	5.0	5.5	6.4	6.5	6.5	6.4	(14)
(15)	Precipitation	PrecAvg	595	50	40	34	30	39	46	69	60	54	57	61	50	(15)
(16)		PrecMax	833	91	85	61	80	69	81	136	157	120	107	111	118	(16)
(17)		PrecMin	442	11	3	9	7	10	8	8	13	14	22	18	15	(17)
(18)		PrecStd	119	23	22	14	19	16	22	33	38	29	23	26	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	2.6	3.6	10.7	22.5	24.2	27.2	24.3	18.1	11.5	7.4	4.0	(19)
(20)			MCWB	1.2	1.2	1.3	5.8	14.6	16.8	20.0	18.9	14.4	10.1	6.5	3.3	(20)
(21)		2%	DB	1.1	1.5	2.2	7.2	18.3	20.8	24.7	21.9	16.2	10.6	6.6	2.9	(21)
(22)			MCWB	0.7	0.7	0.8	3.7	12.5	15.1	19.0	18.1	13.7	9.4	6.1	2.1	(22)
(23)		5%	DB	0.6	0.7	1.5	4.9	15.4	18.6	22.7	20.4	15.2	9.8	5.3	2.3	(23)
(24)			MCWB	0.2	0.2	0.5	2.4	10.8	14.1	18.4	16.9	13.4	8.7	4.6	1.6	(24)
(25)		10%	DB	-0.1	-0.1	0.7	3.4	12.4	17.0	21.2	19.1	14.6	9.0	4.3	1.7	(25)
(26)			MCWB	-0.3	-0.4	0.0	1.6	8.9	13.5	17.6	16.2	13.0	7.9	3.7	1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	1.6	1.7	6.8	15.8	18.3	21.4	20.5	15.8	10.6	6.9	3.5	(27)
(28)			MCDWB	1.7	2.4	2.7	9.5	20.8	21.9	24.3	23.1	17.3	11.1	7.0	3.9	(28)
(29)		2%	WB	0.9	0.9	1.1	4.0	13.1	16.3	20.3	18.9	14.5	9.8	6.2	2.3	(29)
(30)			MCDWB	1.1	1.3	1.8	6.7	17.9	19.6	23.4	21.3	15.5	10.4	6.6	2.6	(30)
(31)		5%	WB	0.3	0.3	0.6	2.7	11.0	15.1	19.1	17.5	13.9	8.9	4.7	1.8	(31)
(32)			MCDWB	0.5	0.7	1.2	4.2	14.4	17.7	22.3	19.6	15.0	9.7	5.2	2.1	(32)
(33)		10%	WB	-0.3	-0.4	0.1	2.0	9.5	14.0	17.8	16.5	13.2	8.0	3.8	1.2	(33)
(34)			MCDWB	0.0	0.0	0.7	3.1	12.2	16.3	20.7	18.5	14.5	9.0	4.3	1.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	6.6	7.3	6.0	5.7	5.0	5.2	4.6	3.9	3.5	3.8	4.7	(35)
(36)			MCDBR	4.7	4.5	5.6	7.5	9.5	7.5	7.3	6.4	4.4	2.8	3.2	3.2	(36)
(37)			MCWBR	4.6	4.1	4.8	5.0	5.3	4.3	4.2	3.8	3.2	2.7	3.2	3.1	(37)
(38)		5% WB	MCDWB	4.8	4.7	5.2	6.2	8.7	6.3	6.1	4.8	3.6	2.7	3.1	3.1	(38)
(39)			MCWBR	4.8	4.4	4.5	4.4	5.4	4.1	4.1	3.5	2.9	2.7	3.2	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.207	0.273	0.303	0.339	0.346	0.349	0.367	0.351	0.328	0.304	0.343	0.355	(40)	
(41)		taud	1.943	2.088	2.139	2.192	2.413	2.482	2.454	2.486	2.529	2.347	2.564	2.603	(41)	
(42)		Ebn at Noon	251	581	753	819	859	864	833	812	752	565	240	79	(42)	
(43)		Edh at Noon	20	58	89	109	98	94	94	83	64	46	19	10	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.08	0.57	2.04	3.79	4.94	5.54	5.15	3.76	2.01	0.72	0.14	0.02	(44)	
(45)		RadStd	0.01	0.11	0.21	0.30	0.45	0.45	0.42	0.45	0.34	0.21	0.12	0.02	0.00	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page